Muhakem Legal AI

Arabic Legal Intelligence for Research, Drafting, and Document Review

---

Project Summary

A research and software prototype for Egyptian legal workflows.

Prepared by Heba Reda Mohamed Omran, Manal Gamil Hassan Alaaeldin, Omnia Mahfouz Abdellatif Mostafa, Nehal Hammam Hassan Abdelhamed, and Mariam Magdy Abdo Mohamed Elatbany.

Graduation Project Advisor: Prof. Ghazal Abdelaty Fahmy

August 2026

1 1. Problem Statement

Legal professionals in Egypt often search across disconnected sources to find the statutory provisions, judicial rulings, and legal commentary relevant to a particular matter. Manual searching is time-consuming and can cause important references to be overlooked. The problem becomes more serious when a general-purpose language model produces unsupported article numbers, fabricated quotations, unrelated arguments, or content from the wrong legal domain.

Formal legal outputs add further requirements. A contract must contain complete and internally consistent clauses. A defense memorandum must reflect the facts and crime classification of the specific case. A legal consultation may require a direct statutory answer, or—when based on case facts—a combination of provisions, judicial rulings, and commentary. A useful system must therefore organize legal knowledge, route different requests to specialized workflows, retrieve relevant evidence, and present structured outputs that remain reviewable by a legal professional.

The core problem addressed by Muhakem is the lack of an Arabic-first, Egypt-focused workspace that combines legal research with practical drafting and document analysis while reducing unsupported claims and preserving the human lawyer's responsibility for the final decision.

2 2. Proposed Solution

Muhakem is a multi-agent legal-AI workspace that routes each request to a specialized legal path: legal research and consultation, contract generation and review, or defense-memorandum generation. Retrieval-Augmented Generation (RAG) grounds model outputs in Egyptian legal material, while hybrid lexical and semantic retrieval improves the identification of relevant passages.

The consultation path combines retrieved statutory articles with related rulings and commentary to support source-aware answers. The contract-generation path retrieves relevant clause precedents, produces a structured editable contract, and enables clause-level modification after generation. The defense-memorandum path structures free-text case facts, classifies the crime type, retrieves crime-relevant legal material, and produces a mandatory multi-section memorandum. Validation layers examine structure, citations, and domain consistency before the result is shown.

The solution is designed to assist rather than replace legal professionals. Its principal value is a traceable workflow: the user can inspect the retrieved references, refine individual clauses or sections, and retain the final professional judgment.

3 3. Technology and System Design

The system uses a React and Vite frontend connected to a FastAPI backend. A router agent directs incoming requests to the appropriate workflow. Qdrant stores searchable legal chunks and metadata; hybrid retrieval combines semantic vector search with lexical/BM25 search. A knowledge-graph layer can connect legal articles to rulings, commentary, and related provisions.

For generation, the project supports Qwen3 served through Ollama as well as compatible online model providers, allowing Online, Hybrid, and Local deployment modes. Muffakir_Embedding_V2 is used for Arabic legal embeddings where configured. Contract review uses OCR and clause-level analysis, while document outputs can be exported as editable DOCX files. SQLite and local authentication are available for a single-user desktop-oriented deployment.

The system records operation-level model usage, including prompt tokens, completion tokens, total tokens, request count, model name, and latency. This supports reproducible academic comparison between local inference and online API deployment.

Layer	Role
Frontend	React/Vite workspace for consultation, drafting, review, and usage analytics
Orchestration	Router and specialized workflow pipelines
Retrieval	Qdrant hybrid search, embeddings, BM25, and optional graph relations
Generation	Qwen3/Ollama or compatible online provider
Document processing	OCR, chunking, clause analysis, validation, and DOCX export

4 4. Project Team

Muhakem was developed as a collaborative graduation project by a five-member team within the Digilians 9 Months Diploma in AI and Data Science. The team members are Heba Reda Mohamed Omran, Manal Gamil Hassan Alaaeldin, Omnia Mahfouz Abdellatif Mostafa, Nehal Hammam Hassan Abdelhamed, and Mariam Magdy Abdo Mohamed Elatbany.

The project was developed under the supervision of Prof. Ghazal Abdelaty Fahmy, with the academic and training context of the Digilians initiative. Team responsibilities covered legal-domain research, data preparation, retrieval, model integration, backend services, frontend experience, document generation, testing, and evaluation.

5 5. Market Analysis

Muhakem targets legal consultants, law offices, legal researchers, corporate legal departments, and legal-education or professional-training environments that work primarily in Arabic and Egyptian legal materials. These users share three needs: faster access to relevant legal sources, structured drafting support, and a reviewable trail from a generated answer to the material used to support it.

The market opportunity is shaped by a gap between general-purpose AI assistants and specialized legal workflows. General assistants can be fluent but may not provide reliable Egyptian-law retrieval, clause structure, or domain-specific validation. Traditional legal research tools may provide documents but often do not unify conversational research, drafting, clause editing, and document review in one Arabic-first workspace. Muhakem differentiates itself through specialized routing, source-grounded retrieval, clause-level interaction, and deployment flexibility.

The initial go-to-market focus should be professional pilots with small and medium-sized law offices and legal departments. Early evaluation should measure retrieval relevance, citation support, drafting time saved, correction rate, latency, and user trust. The project should avoid claiming legal accuracy without expert review; validation by practicing lawyers is a core part of responsible market adoption.

6 6. Business Plan

Muhakem can begin as a professional productivity platform with a pilot-led model. The first stage is a controlled pilot for legal professionals using fictional or approved anonymized documents. The second stage is a subscription product for individual professionals and small offices, with workspace features, document quotas, source packs, and usage analytics. The third stage can offer organization plans with private deployment, access controls, audit logs, and domain-specific legal collections.

A practical revenue model combines recurring subscriptions with optional professional onboarding and private-deployment services. Pricing should be validated through pilot interviews rather than fixed prematurely. The online version can be priced per workspace or user tier, while a local or hybrid edition can be offered where privacy, predictable infrastructure, or institutional policy makes local inference valuable. The cost model should distinguish provider API charges from local compute, storage, support, and maintenance.

The principal business risks are legal-domain reliability, changing model costs, data confidentiality, and user overreliance on generated text. Mitigations include source display, human approval, citation checks, anonymized evaluation data, clear disclaimers, configurable deployment modes, and continuous testing against representative Egyptian legal workflows.

7 Conclusion

Muhakem addresses a practical and research-relevant problem: how to make Arabic legal AI more useful, structured, and reviewable without removing the legal professional from the decision loop. By combining specialized workflows, hybrid retrieval, source-aware generation, validation, and deployment flexibility, the project provides a foundation for a responsible legal-assistance product and for continued evaluation with real legal practitioners.

Disclaimer. Muhakem is a research and software prototype. Its outputs are not a substitute for advice from a qualified lawyer and should be checked against applicable legislation and official sources.